

The Influence of Plant Partner Variability on Population Dynamics and Network Roles in Willamette Valley Prairies

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I. Introduction

Species-level traits are a central focus in ecology because they influence species fitness (Bergamo, Traveset, and Lázaro 2021), regulate population dynamics (Herrera 2005), and even shape evolutionary trajectories (Cosmo et al. 2023). These traits structure interactions within plant–pollinator networks, which is important because mutualistic networks provide essential ecosystem services in both natural and agricultural systems (Waser 2006). Network theory predicts that pairwise interactions change through space. Highly dynamic pairwise interactions complicates our ability to generalize plant species' network roles across a landscape. Examining partner variability as a plant trait, and how this trait influences network niche through space can help inform conservation planning of rare, cultural, and restoration valuable plants.

Mutualistic networks consist of two interacting communities called nodes that are connected by links. Using a bipartite approach, we can map species interactions and identify species roles within a network. Well-characterized pairwise interactions change spatially and temporally (L. W. Aarssen 1989; L. Aarssen 1992; Bengtsson, Fagerström, and Rydin 1994; Carstensen et al. 2014). This variation is driven by temporal variation in species composition and phenology (Alarcón, Waser, and Ollerton 2008; Olesen et al. 2008) and interaction re-wiring due to disturbance (Ponisio 2020) or resource availability (Valdovinos et al. 2016).

An important aspect of species diversity is beta diversity—the spatial turnover of species interactions (Carstensen et al. 2014). The dissimilarity in beta-diversity of populations in a region is low if populations are interacting with the same species compositions and high when interacting with unique compositions (Ponisio 2020). Plant species exhibit substantial variability in their interaction partners among conspecific individuals within a population (Gómez, Perfectti, and Jordano 2011), across populations (González, Murúa, and Pérez 2015), and among communities (Herrera 2005). Flexible pairwise interactions likely contribute to a plant species ability to shift between network niches when community composition changes. Thus, it is valuable to examine whether species that are flexible in their interaction partners can occupy different as community composition changes through space.

This study focuses on prairies in the Willamette Valley ecoregion in Oregon. Over the past 175 years native prairie ecosystems in the Pacific Northwest have suffered a drastic decline, with approximately 90-98% lost throughout the region (Sinclair et al. 2006; Altman 2011). Advancing our understanding of interaction-based ecosystem services by treating partner variability as a trait and evaluating its impacts on population dynamics

and network niches is essential for informing conservation planning of these imperiled ecosystems. Yet, studies that link partner variability and network niche remains scarce (see Ponisio (2020)). Here, I assess plant partner variability as a trait, and how this trait may predict population dynamics and network niches.

Across populations plants show substantial variation in their interaction partners. This is likely explained by shifting compositions of available interaction partners as community compositions change with geographic distance (Carstensen et al. 2014). As plants become more flexible in their interaction partners, their interannual population dynamics are likely to be less variable because they can more reliably obtain necessary pollination services. Moreover, the network roles they occupy may shift accordingly. This flexible behavior has been observed in plant populations of *L. latifolia* (Herrera 2005), and pollinator populations (Ponisio 2020; Gaiarsa, Kremen, and Ponisio 2021). These insights have motivated three hypothesis: (1) As beta-diversity increases, the role of a plant within the network will shift. (2) As plants become more flexible in their interaction partners they will show more variation in their network niche (3) Plants that are more flexible in their interaction partners will have a negative relationship with changes in floral abundance among years.

II. Data

Data was provided by our collaborator Susan Waters with Quamash EcoResearch. This dataset included plant-pollinator interactions across 10 sites within the Willamette Valley Ecoregion in Oregon from 2019-2022. The data was collected using standardized timed hand-netting survey methods where bees are collected when actively engaging in pollination. The interacting plant species partners were recorded in all observations and identified to species. Bee species were identified as morphospecies or species when feasible. Site and date (survey round) were recorded for all observations. Using the raw data I created a vegetation dataset that included all plant species sampled and their abundances for each survey round within a site and year, a network dataset that included all plant pollinator interactions, and a spatial dataset with the latitude and longitude converted to a vector (see below). The preparation of datasets and the development of the analytical approach was based on the steps used in Ponisio (2020). The code was sourced from github and modified to meet the analytical destination in the present study.

III. Data Wrangling

All data preparation was done in 0DataPrep.R

```
setwd('C:/pisgah-prairie/data')  
#Loads packages  
source('../analysis/src/packages.R')
```

Specimen Data

To clean the desired specimen dataset that includes all plant-pollinator interactions I used the following approach. First, I changed the date to a single format and creating a year column. Vegetation data was standardized using GBIF. All empty cells and NA values in the bee genus, bee species and plant genus, plant species columns were dropped. I used the `sum(is.na())` as a built in test for the previous step. I created a genus

species column for bees and an interactions column for all interactions of bees and plants. I then subsetting the data to only include bee families. This resulting dataset was used as a specimen dataset.

```
setwd('C:/piscgah-prairie/data')
source('../data/src/misc.R')
source('../data/src/prepNets.R')

#####
## Clean specimen data
#####
spec <- read.csv(
  "raw/susans_data_normalized.csv",
  colClasses = "character")

dim(spec) #14211 x 11
```

```
## [1] 14211    19
```

```
#changes date to single format, date class, and create year column
spec$Date <- lubridate::parse_date_time(
  spec$Date,
  orders = c("d-b-y", "Ymd")
)

spec$Date <- as.Date(spec$Date)

class(spec$Date)
```

```
## [1] "Date"
```

```
spec <- spec %>%
  dplyr::mutate(
    Year = year(Date))

#drop empty cells and NA
spec <- spec[spec$Genus != '',]
spec <- spec[spec$Species != '',]
spec <- spec[spec$PlantGenusSpecies != '',]
spec <- spec[!is.na(spec$Genus),]
spec <- spec[!is.na(spec$Species),]
spec <- spec[!is.na(spec$PlantGenusSpecies),]

# Count NAs in each column
sum(is.na(spec$Genus))
```

```
## [1] 0
```

```
sum(is.na(spec$Species))
```

```
## [1] 0
```

```
sum(is.na(spec$PlantGenusSpecies))
```

```
## [1] 0
```

```
#Create genus species column
spec$GenusSpecies <- paste(spec$Genus,
                           spec$Species)

#interactions
spec$Int <- paste(spec$GenusSpecies,
                  spec$PlantGenusSpecies)

dim(spec) #5147 22
```

```
## [1] 5152 22
```

```
## drop non-bees
unique(spec$Family)
```

```
## [1] "Andrenidae" "Apidae" "Bombyliidae" "Conopidae"
## [5] "Halictidae" "Chrysomelidae" "Coccinellidae" "Rhagionidae"
## [9] "Syrphidae" "Vespidae" "Megachilidae" "Cleridae"
## [13] "Colletidae" "\tHalictidae" "Pompilidae" "Crabronidae"
## [17] "Cerambycidae" "Attelabidae" "" "Pollenidae"
## [21] "Tachinidae" "Meloidae" "Calliphoridae" "\tSyrphidae"
## [25] "Chrysididae" "Elateridae" "Sphecidae"
```

```
bee.fams <- c("Halictidae", "Andrenidae", "Apidae", "Megachilidae",
             "Colletidae", "Bombyliidae")
```

```
spec <- spec[spec$Family %in% bee.fams,]
```

```
#summary
print(paste("Bee species", length(unique(spec$GenusSpecies))))
```

```
## [1] "Bee species 123"
```

```
print(paste("Plant species", length(unique(spec$PlantGenusSpecies))))
```

```
## [1] "Plant species 89"
```

```
print(paste("Bee genera", length(unique(spec$Genus))))
```

```
## [1] "Bee genera 23"
```

```
print(paste("Interactions", length(unique(spec$Int))))
```

```
## [1] "Interactions 623"
```

```
print(paste("Specimens", nrow(spec)))
```

```
## [1] "Specimens 2885"
```

```
print(length(unique(spec$Site)))
```

```
## [1] 10
```

```
dim(spec)
```

```
## [1] 2885 22
```

Network Data

To prepare the network matrices for network analysis, I formatted the specimen dataset into a list of interaction matrices for each survey round at each site within a year.

```
##*****
## networks
##*****
spec <- spec[!spec$PlantGenusSpecies == "",]

nets <- break.net(spec)
nets <- unlist(nets, recursive=FALSE)
```

Vegetation Data

To create a vegetation dataset with floral counts, I standardized the plant naming conventions using GBIF. I renamed the columns to match the specimen dataset and changed the date to a single format and created a year column. I then dropped all empty cells and NA values in the plant genus and species columns.

```

#####
## clean veg data
#####

##Create veg dataset
raw <- read_csv('C:/pisgah-prairie/data/raw/PPP_PollObs_OR_upto2022_NEW_IDS_20240108 (2).csv')
norm1 <- read_csv('C:/pisgah-prairie/data/raw/normalized1.csv')
norm2 <- read_csv("C:/pisgah-prairie/data/raw/normalized2.csv")
norm3 <- read_csv('C:/pisgah-prairie/data/raw/normalized3.csv')

norm <- rbind(norm1, norm2, norm3)

raw <- raw %>%
  select(State, County, Latitude, Longitude, site, date, time, temp, hum, wind, plants_per_m2, patch_size)

veg <- cbind(norm, raw)

#select and rename columns
veg <- veg %>%
  select(-occurrenceId, -acceptedUsageKey, -usageKey, -authorship, -canonicalName, -verbatimScientificName, -matchType, -confidence, -key) %>%
  rename(Date = date, Site = site, PlantGenusSpecies = species, Genus = genus, Family = family, Kingdom = kingdom, Phylum = phylum, Class = class, Order = order, FlowerNum = plants_per_m2)

#changes date to single format, date class, and create year column
veg$Date <- lubridate::parse_date_time(
  veg$Date,
  orders = c("d-b-y", "Ymd")
)

spec$Date <- as.Date(spec$Date)

class(spec$Date)

## [1] "Date"

```

```
#create year column
spec <- spec %>%
  dplyr::mutate(
    Year = year(Date))

#drop empty cells and NA
veg <- veg[veg$Genus != '',]
veg <- veg[veg$PlantGenusSpecies != '',]
veg <- veg[!is.na(veg$Genus),]
veg <- veg[!is.na(veg$PlantGenusSpecies),]

## correct date format
veg$Date <- as.Date(veg$Date, format='%m/%d/%y')
veg$Year <- format(veg$Date, format='%Y')
veg$doy <- as.numeric(strftime(veg$Date, format='%j'))

#identify species occurrences and convert to binary
veg$FlowerNum[is.na(veg$FlowerNum)] <- 0
veg$Occ <- veg$FlowerNum
veg$Occ[veg$Occ > 0] <- 1
```

IV. Observed & Null Communities

I created null and observed community interaction matrices (1nulls. R). I generated randomized communities to standardize partner beta diversity scores. I constrained null models using alpha diversity for the beta diversity calculation because species vary in their partner richness across populations within the landscape. I first created observed communities for each site and year and then repeated this step to create null communities using modified methods proposed by Vázquez et al. (2007).


```
setwd('C:/pisgah-prairie/analysis')
this.script <- "nulls"
source('src/misc.R')
source('src/calcPca.R')
source('src/calcSpec.R')
source('src/commPrep.R')
source('src/vaznull2.R')

load('../data/specimens/spec1.Rdata')
load('../data/specimens/nets1.Rdata')
save.path <- 'saved'

save.dir.comm <- "saved/communities"
save.dir.nulls <- "saved/nulls"

type <- "plant"
nnull <- 9
species.type="PlantGenusSpecies"
species.type.int="GenusSpecies"

sites <- unique(spec$Site)
years <- unique(spec$Year)

## *****
## year by species matrices plants
## *****
comms <- lapply(years, calcSiteBeta,
                species.type=species.type,
                spec=spec,
                species.type.int=species.type.int)

comm <- makePretty(comms, spec)

## *****
## alpha div nulls
## *****
nulls <- rapply(comm$comm, vaznull.2, N=nnull, how="replace")

## *****
```

```
## occurrence nulls
## *****
occ.null <- function(web){
  simulate(vegan::nullmodel(web, method="quasiswap"),1)[,1]
}

rep.occ.null <- function(web, N){
  replicate(N, occ.null(web), simplify = FALSE)
}

nulls <- rapply(comm$comm, rep.occ.null, N=nnull, how="replace")
```

V. Beta Diversity

To quantify how plants are flexible in their interaction partners across populations, I calculated Beta diversity—the change in interaction partners among plant populations across space within a single year (2partner.R). Using the observed communities and the null communities I calculated beta diversity constrained by alpha diversity using bipartite:betadisper, and then quantified the dissimilarity across populations within a year using the Chao method. Beta-diversity was only calculated for species that were observed at least three times within a single year. The output dataset included beta-diversity values for each species at a site within a single year, and these values were used to represent plant partner variability.

```

setwd('C:/pisgah-prairie/analysis')
source('src/initialize_beta.R')
source('src/misc.R')
source('src/calcPca.R')
source('src/calcSpec.R')
source('src/beta.R')

## metrics for taking the average and variability of beta div
var.method <- cv

dis <- mapply(function(a, b, c, d)
  calcBeta(comm= a, ## observed communities
            dis.method, ## dissimilarity metric
            nulls=b, ## null communities
            occ=binary, ## binary weighted?
            sub=type,
            zscore=FALSE), ## used chase method
  a=comm$comm,
  b= nulls,
  SIMPLIFY=FALSE)

beta.dist <- makeBetaDataPretty()

head(beta.dist, n = 10)

```

##	Year	PlantGenusSpecies	dist	Site	Date
## 1	2019	Camassia quamash	0.3562303	Fisher Butte	2019-04-23
## 2	2019	Camassia quamash	0.3039737	Fisher Butte	2019-05-13
## 3	2019	Camassia quamash	0.1090871	Jefferson Farm	2019-04-24
## 4	2019	Camassia quamash	0.6163603	Jefferson Farm	2019-05-17
## 5	2019	Camassia quamash	0.3231099	Oxbow West	2019-04-25
## 6	2019	Clarkia purpurea	0.2500000	Bald Hill	2019-07-08
## 7	2019	Clarkia purpurea	0.4000000	Finley Wildlife Refuge	2019-06-13
## 8	2019	Clarkia purpurea	0.1500000	Jefferson Farm	2019-07-05
## 9	2019	Crepis setosa	0.3715583	Bald Hill	2019-06-24
## 10	2019	Crepis setosa	0.3590110	Jefferson Farm	2019-06-19

```
# 1. Compute mean for each site
site_means <- beta.dist %>%
  group_by(Site) %>%
  summarise(mean_dist = mean(dist, na.rm = TRUE))

#visualize beta diversity
partHistPlot <- ggplot2::ggplot(beta.dist, aes(x = dist)) +
  geom_histogram(fill = "blue", color = "white") +
  geom_vline(data = site_means,
             aes(xintercept = mean_dist),
             color = "red", linewidth = 1.2) +
  labs(x = "Partner Variability", y = "Frequency") +
  theme_classic() +
  theme(text = element_text(size = 30)) +
  facet_wrap(~Site, ncol = 3)
```

VI. Network Niche Variability

I calculated plant network roles and a principal component analysis (PCA) to later be used as my response variables regressed against partner variability (3pca.R). I used the network and specimen dataset and applied `bipartite::specieslevel` function to calculate pollinator and plant network roles. I subsetting the data to only include plant species roles and dropped pollinators. I selected PCA loadings of interest which encompass a species network role (i.e., rare degree, weighted betweenness, weighted closeness, niche overlap, species strength,) and calculated the mean and variance of the PCA scores. The output dataset was a list of PCA scores for each plant species at a site within a single year. I chose to use the variance of the PCA scores to represent a plant's ability to occupy different network niches across populations in the landscape because variance better captures the spread of the data than the would otherwise be lost if using the mean.

```
setwd('C:/pisgah-prairie/analysis')
source('src/misc.R')
source('src/calcPca.R')
source('src/calcSpec.R')
source('src/commPrep.R')
source('src/vaznull2.R')

load('../data/specimens/spec1.Rdata')
load('../data/specimens/nets1.Rdata')
this.script <- "role"
type <- "all"

## applies specieslevel from bipartite to networks
## calculates the species roles from a network and returns a dataframe
species.roles <- calcSpec(nets, spec, dist.metric="chao")
species.roles <- species.roles[species.roles$speciesType == "plant",]

head(species.roles, n = 10)
```

```

##      degree normalised.degree species.strength interaction.push.pull nestedrank
## 11      5      0.5000000      4.256410      0.6512821      0.00
## 12      4      0.4000000      1.910256      0.2275641      1.00
## 13      5      0.5000000      3.833333      0.5666667      0.50
## 22      1      0.1250000      1.000000      0.0000000      0.75
## 23      1      0.1250000      1.000000      0.0000000      1.00
## 24      3      0.3750000      2.750000      0.5833333      0.00
## 25      2      0.2500000      1.250000      0.1250000      0.50
## 26      2      0.2500000      2.000000      0.5000000      0.25
## 34      1      0.1428571      1.000000      0.0000000      0.75
## 35      2      0.2857143      1.333333      0.1666667      0.25
##      PDI resource.range species.specificity.index      PSI
## 11 0.9629630      0.5555556      0.7288690 4.256410
## 12 0.6666667      0.6666667      0.4082483 1.910256
## 13 0.7777778      0.5555556      0.3685139 3.833333
## 22 1.0000000      1.0000000      1.0000000 1.000000
## 23 1.0000000      1.0000000      1.0000000 1.000000
## 24 0.8095238      0.7142857      0.5479886 2.750000
## 25 0.8571429      0.8571429      0.6546537 1.250000
## 26 0.9285714      0.8571429      0.7014724 2.000000
## 34 1.0000000      1.0000000      1.0000000 1.000000
## 35 0.8333333      0.8333333      0.6454972 1.333333
##      node.specialisation.index.NSI betweenness weighted.betweenness closeness
## 11      1      0      0 0.3333333
## 12      1      0      1 0.3333333
## 13      1      0      0 0.3333333
## 22      NaN      0      0 0.0000000
## 23      NaN      0      0 0.0000000
## 24      1      0      0 0.5000000
## 25      1      0      0 0.5000000
## 26      NaN      0      0 0.0000000
## 34      NaN      0      0 0.0000000
## 35      1      0      0 0.3333333
##      weighted.closeness Fisher.alpha partner.diversity effective.partners
## 11      0.4369748 2.496744e+00      0.9089087      2.481613
## 12      0.2857143 2.147484e+09      1.3862944      4.000000
## 13      0.1904762 1.411633e+01      1.5607104      4.762203
## 22      0.0000000 1.342178e+08      0.0000000      1.000000
## 23      0.0000000 1.342178e+08      0.0000000      1.000000

```

```

## 24      1.5000000 1.988798e+00      1.0042425      2.729839
## 25      0.5000000 2.684355e+08      0.6931472      2.000000
## 26      0.0000000 2.622303e+00      0.6365142      1.889882
## 34      0.0000000 1.342178e+08      0.0000000      1.000000
## 35      0.5000000 2.684355e+08      0.6931472      2.000000
##      proportional.generality proportional.similarity      d tot.int      mean.k
## 11      0.4346074      0.67788462 0.6766942      16 4.666667
## 12      0.7005241      0.48076923 0.2129318      4 4.666667
## 13      0.8340095      0.34615385 0.7397181      6 4.666667
## 22      0.1482974      0.07142857 1.0000000      1 1.800000
## 23      0.1482974      0.07142857 1.0000000      1 1.800000
## 24      0.4048281      0.57142857 0.7284356      7 1.800000
## 25      0.2965949      0.35714286 0.4421141      2 1.800000
## 26      0.2802646      0.21428571 1.0000000      3 1.800000
## 34      0.1655218      0.09090909 1.0000000      1 1.800000
## 35      0.3310437      0.36363636 0.5000000      2 1.800000
##      sd.k      k niche.overlap rare.degree      GenusSpecies
## 11 0.5773503 0.5773503 0.6566339      11.0 Erigeron decumbens
## 12 0.5773503 -1.1547005 0.4144278      10.0 Eriophyllum lanatum
## 13 0.5773503 0.5773503 0.6882492      13.0 Potentilla gracilis
## 22 0.8366600 -0.9561829 1.0000000      1.0 Castilleja levisecta
## 23 0.8366600 -0.9561829 1.0000000      1.0 Heracleum maximum
## 24 0.8366600 1.4342743 0.9127907      3.0 Potentilla gracilis
## 25 0.8366600 0.2390457 0.9127907      3.0 Sidalcea asprella
## 26 0.8366600 0.2390457 1.0000000      2.5 Sidalcea campestris
## 34 0.8366600 -0.9561829 1.0000000      1.0 Brassica rapa
## 35 0.8366600 0.2390457 0.7613636      3.0 Heracleum maximum
##      speciesType      Site      Date Year
## 11      plant Ankeny Wildlife Refuge 2019-06-13 2019
## 12      plant Ankeny Wildlife Refuge 2019-06-13 2019
## 13      plant Ankeny Wildlife Refuge 2019-06-13 2019
## 22      plant Ankeny Wildlife Refuge 2021-05-13 2021
## 23      plant Ankeny Wildlife Refuge 2021-05-13 2021
## 24      plant Ankeny Wildlife Refuge 2021-05-13 2021
## 25      plant Ankeny Wildlife Refuge 2021-05-13 2021
## 26      plant Ankeny Wildlife Refuge 2021-05-13 2021
## 34      plant Ankeny Wildlife Refuge 2021-05-15 2021
## 35      plant Ankeny Wildlife Refuge 2021-05-15 2021
##      SpSiteYear

```

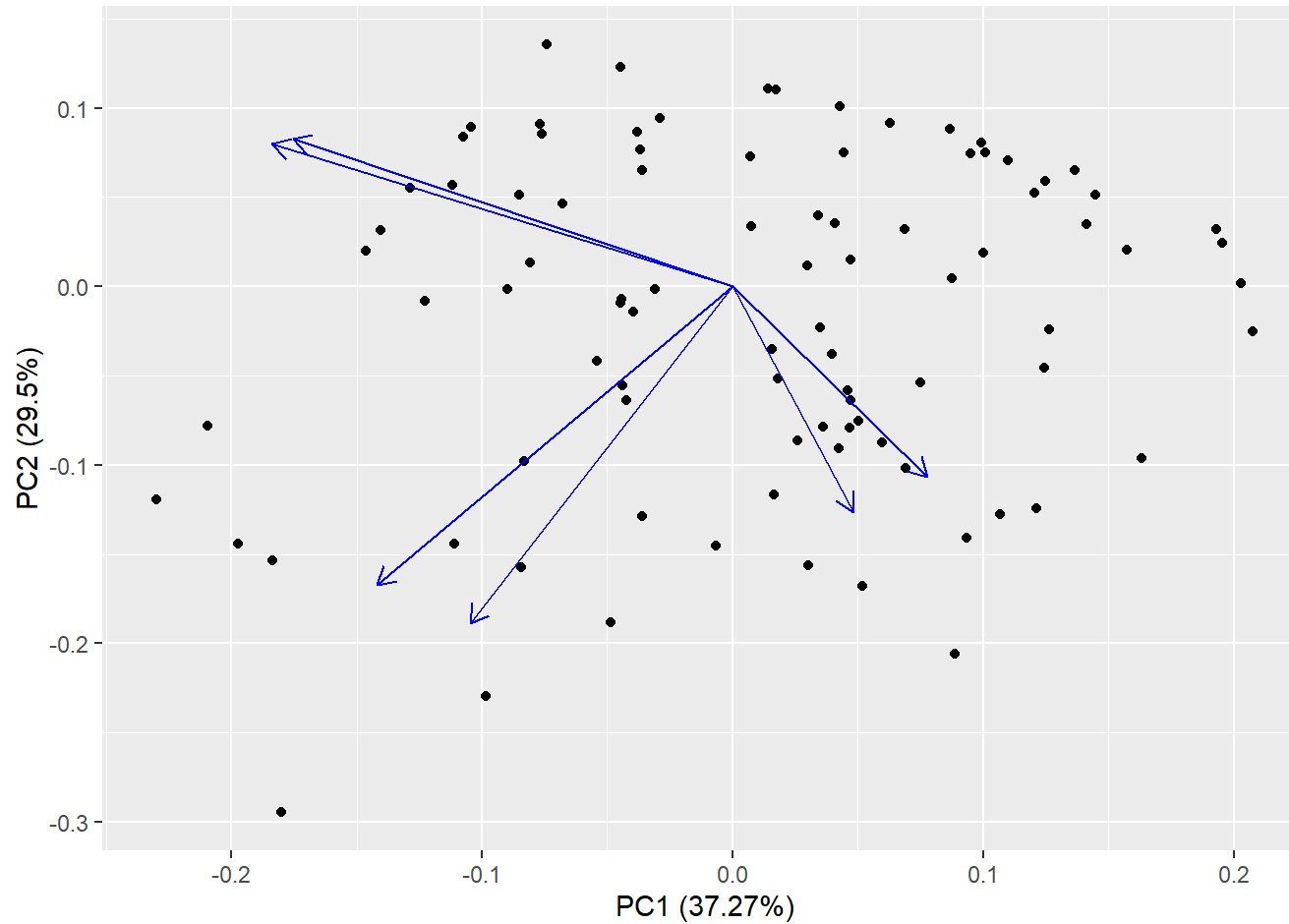
```
## 11 Erigeron decumbensAnkeny Wildlife Refuge2019
## 12 Eriophyllum lanatumAnkeny Wildlife Refuge2019
## 13 Potentilla gracilisAnkeny Wildlife Refuge2019
## 22 Castilleja levisectaAnkeny Wildlife Refuge2021
## 23 Heracleum maximumAnkeny Wildlife Refuge2021
## 24 Potentilla gracilisAnkeny Wildlife Refuge2021
## 25 Sidalcea asprellaAnkeny Wildlife Refuge2021
## 26 Sidalcea campestrisAnkeny Wildlife Refuge2021
## 34 Brassica rapaAnkeny Wildlife Refuge2021
## 35 Heracleum maximumAnkeny Wildlife Refuge2021
```

```
## vector of pca loadings of interest
loadings <- c(1)
metrics <- c("rare.degree",
             "weighted.betweenness",
             "weighted.closeness",
             "niche.overlap",
             "species.strength",
             "d")

## the metrics used in the PCA
var.method <- cv
ave.method <- mean

#changed from pollinator to plant
plant.pca.scores <- calcPcaMeanVar(species.roles=species.roles,
                                   var.method=var.method,
                                   ave.method=ave.method,
                                   metrics= metrics,
                                   loadings=loadings,
                                   agg.col = "Year")

autoplot(plant.pca.scores$'2019'$pca.loadings, loadings=TRUE,
         loadings.colour = 'blue')
```

```
plant.pca.scores.df <- do.call(rbind, lapply(plant.pca.scores, function(x) x$pcas))
```

VII. Floral Abundance

I calculated floral abundance using the vegetation dataset (4detlaAbund.R). I first calculated floral abundance for each plant species in a site within a single year. I then calculated the difference in abundance between year pairs (e.g., 2019-2020, 2020-2021, etc.), and then the log-ratio of these values. The resulting dataset was the floral abundance dataset which had log-ratio values of the difference in abundance between year pairs.

```
setwd('C:/pisgah-prairie/analysis')
source('src/setup.R')
source('src/delta.R')
source('src/misc.R')
load('../data/specimens/spec1.Rdata')
load('../data/veg/veg1.Rdata')

## Calculate the abundance of each species in each year
plant.abund <- calcPlantABund(spec)

## calculate the difference the log ratio of abundance between year pairs
plant.delta <- calcPlantYearDiff(plant.abund)
```

```
## [1] 2019
## [1] 2020
## [1] 2021
```

```
#converts list to dataframe
plant.delta.df <- bind_rows(plant.delta , .id = "Year")

plant.delta.tidy <- plant.delta.df%>%
  rename(PlantGenusSpecies = GenusSpecies) %>%
  mutate(Site = as.character(Site),
         Year = as.character(Year),
         PlantGenusSpecies = as.character(PlantGenusSpecies))
```

VIII. Join Datasets

I joined datasets that I needed to create my linear models (5joinData.R). The datasets of interest were beta-diversity, PCA scores, network roles, and floral abundance datasets. I standardized class type for year, site, plant genus species as character. I then created a key in each dataset to later join the datasets into a single data frame. The output data frame included beta-diversity values, PCA scores (mean and variance), network roles, and log-ratio floral abundances (for year pairs) for each plant species at a site within a single year.

```
setwd('C:/piscagah-prairie/analysis/saved')
load(' ../saved/results/specRoles.Rdata')
load(' ../saved/results/deltaAbund.Rdata')
load(' ../saved/results/partnerVar.Rdata')
load(' ../saved/results/pcaVar.Rdata')

#Create key SpSiteYear and change class to character for each dataframe
delta.tidy <- plant.delta.tidy %>%
  dplyr::mutate(
    PlantGenusSpecies = as.character(PlantGenusSpecies),
    Year = as.character(Year),
    Site = as.character(Site),
    SpSiteYear = paste0(PlantGenusSpecies, Site, Year, " "),
    SpSiteYear = trimws(SpSiteYear))

beta.dist.tidy <- beta.dist %>%
  dplyr::mutate(
    PlantGenusSpecies = as.character(PlantGenusSpecies),
    Year = as.character(Year),
    Site = as.character(Site),
    SpSiteYear = paste0(PlantGenusSpecies, Site, Year, " "),
    SpSiteYear = trimws(SpSiteYear))

spec.roles.tidy <- species.roles %>%
  dplyr::rename(PlantGenusSpecies = GenusSpecies) %>%
  dplyr::mutate(
    PlantGenusSpecies = as.character(PlantGenusSpecies),
    Year = as.character(Year),
    Site = as.character(Site),
    SpSiteYear = paste0(PlantGenusSpecies, Site, Year, " "),
    SpSiteYear = trimws(SpSiteYear))

pca.scores.tidy <- plant.pca.scores.df %>%
  dplyr::rename(PlantGenusSpecies = GenusSpecies) %>%
  dplyr::mutate(
    PlantGenusSpecies = as.character(PlantGenusSpecies),
    Year = as.character(Year),
    Site = as.character(Site),
    SpSiteYear = paste0(PlantGenusSpecies, Site, Year, " "),
```

```
SpSiteYear = trimws(SpSiteYear))
```

```
#Check structure of SpSiteYear for consistency  
str(delta.tidy$SpSiteYear)
```

```
## chr [1:321] "Camassia quamashFisher Butte2019" ...
```

```
str(beta.dist.tidy$SpSiteYear)
```

```
## chr [1:387] "Camassia quamashFisher Butte2019" ...
```

```
str(spec.roles.tidy$SpSiteYear)
```

```
## chr [1:407] "Erigeron decumbensAnkeny Wildlife Refuge2019" ...
```

```
head(delta.tidy$SpSiteYear, n = 10)
```

```
## [1] "Camassia quamashFisher Butte2019"  
## [2] "Cardamine pendulifloraFisher Butte2019"  
## [3] "Lomatium bradshawiiFisher Butte2019"  
## [4] "Ranunculus alismifoliusFisher Butte2019"  
## [5] "Ranunculus orthorhynchusFisher Butte2019"  
## [6] "Plagiobothrys figuratusFisher Butte2019"  
## [7] "Castilleja tenuisFisher Butte2019"  
## [8] "Eriophyllum lanatumFisher Butte2019"  
## [9] "Ranunculus occidentalisFisher Butte2019"  
## [10] "Downingia yinaFisher Butte2019"
```

```
head(beta.dist.tidy$SpSiteYear, n = 10)
```

```
## [1] "Camassia quamashFisher Butte2019"  
## [2] "Camassia quamashFisher Butte2019"  
## [3] "Camassia quamashJefferson Farm2019"  
## [4] "Camassia quamashJefferson Farm2019"  
## [5] "Camassia quamashOxbow West2019"  
## [6] "Clarkia purpureaBald Hill2019"  
## [7] "Clarkia purpureaFinley Wildlife Refuge2019"  
## [8] "Clarkia purpureaJefferson Farm2019"  
## [9] "Crepis setosaBald Hill2019"  
## [10] "Crepis setosaJefferson Farm2019"
```

```
head(spec.roles.tidy$SpSiteYear, n = 10)
```

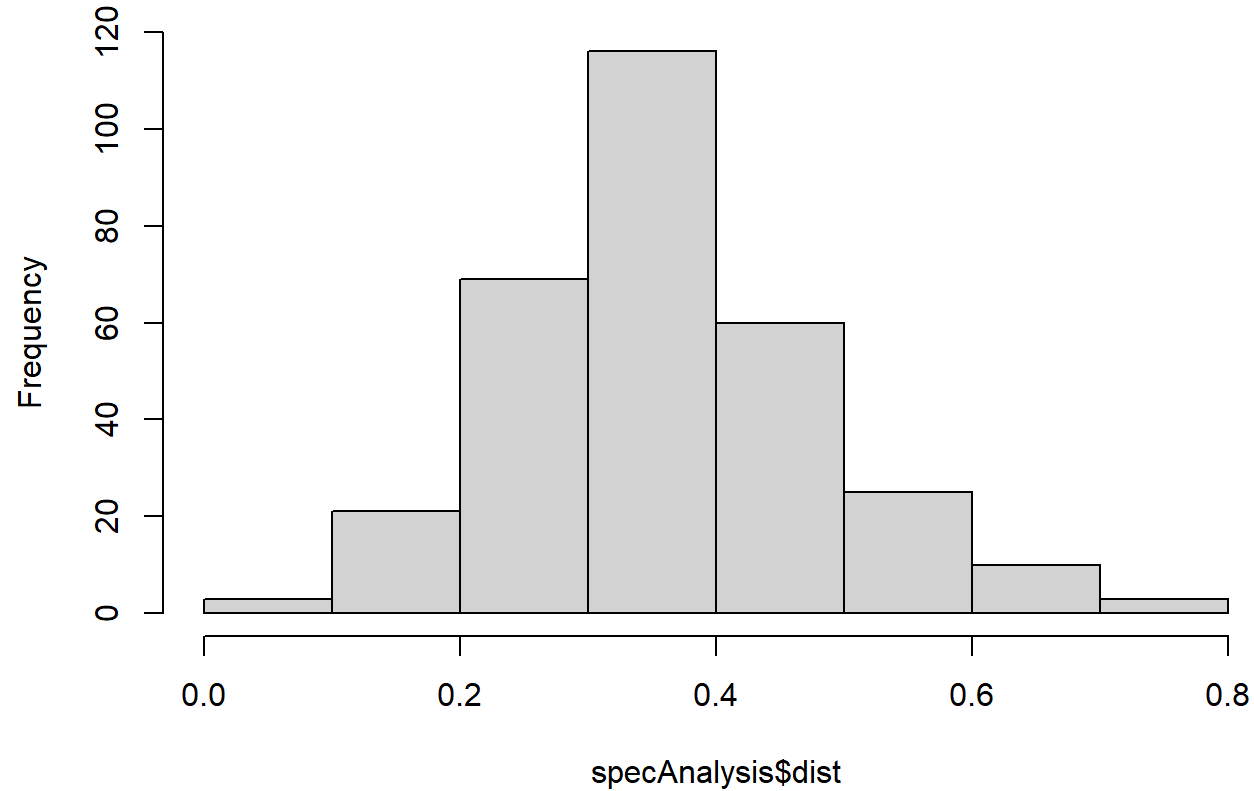
```
## [1] "Erigeron decumbensAnkeny Wildlife Refuge2019"  
## [2] "Eriophyllum lanatumAnkeny Wildlife Refuge2019"  
## [3] "Potentilla gracilisAnkeny Wildlife Refuge2019"  
## [4] "Castilleja levisectaAnkeny Wildlife Refuge2021"  
## [5] "Heracleum maximumAnkeny Wildlife Refuge2021"  
## [6] "Potentilla gracilisAnkeny Wildlife Refuge2021"  
## [7] "Sidalcea asprellaAnkeny Wildlife Refuge2021"  
## [8] "Sidalcea campestrisAnkeny Wildlife Refuge2021"  
## [9] "Brassica rapaAnkeny Wildlife Refuge2021"  
## [10] "Heracleum maximumAnkeny Wildlife Refuge2021"
```

```
#Join dataframes
specAnalysis <- spec.roles.tidy %>%
  dplyr::left_join(delta.tidy %>% select(-Year, -Site, -PlantGenusSpecies), by = "SpSiteYear") %>%
  dplyr::left_join(beta.dist.tidy %>% select(-Year, -Site, -Date, -PlantGenusSpecies), by = "SpSiteYear") %>%
  dplyr::left_join(pca.scores.tidy %>% select(-Year, -Site, -PlantGenusSpecies), by = "SpSiteYear")

# Filter data for non-zero deltaAbund and beta dispersion
specAnalysis <- specAnalysis %>%
  dplyr::filter(dist != 0, deltaAbund != 0, mean.pca1 !=0) %>%
  na.omit()

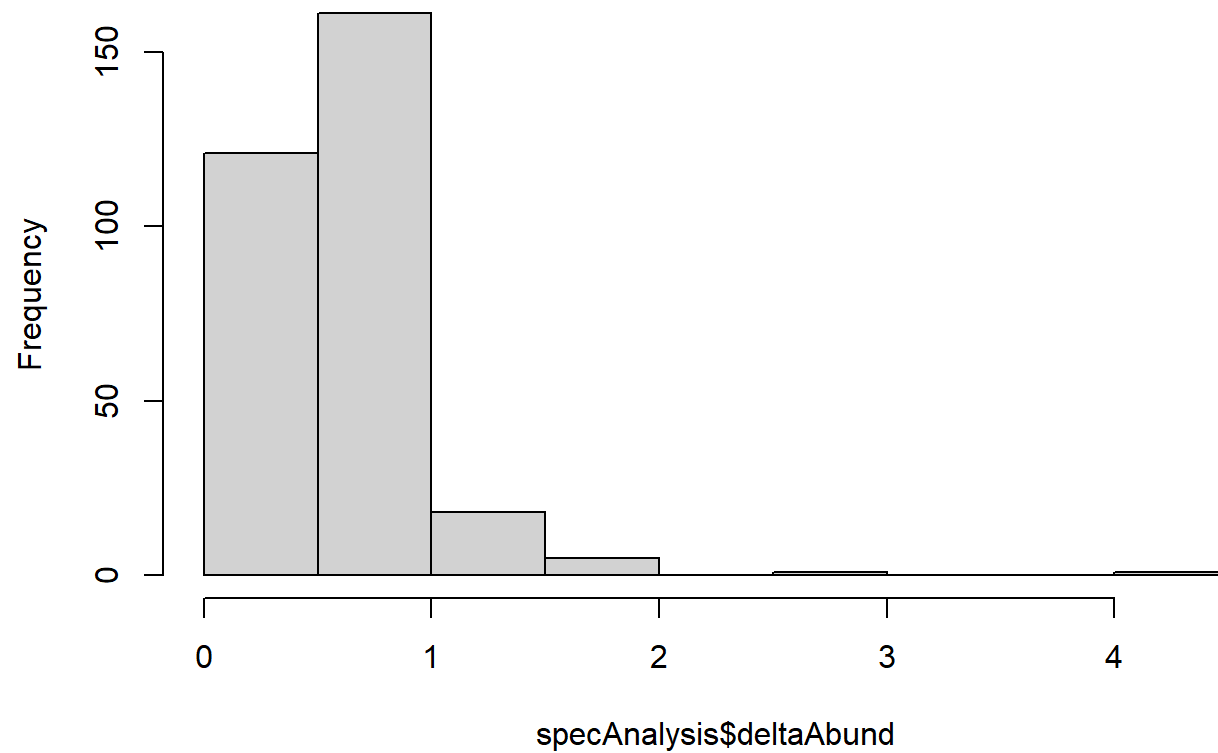
#visualize data of interest
hist(specAnalysis$dist)
```

Histogram of specAnalysis\$dist



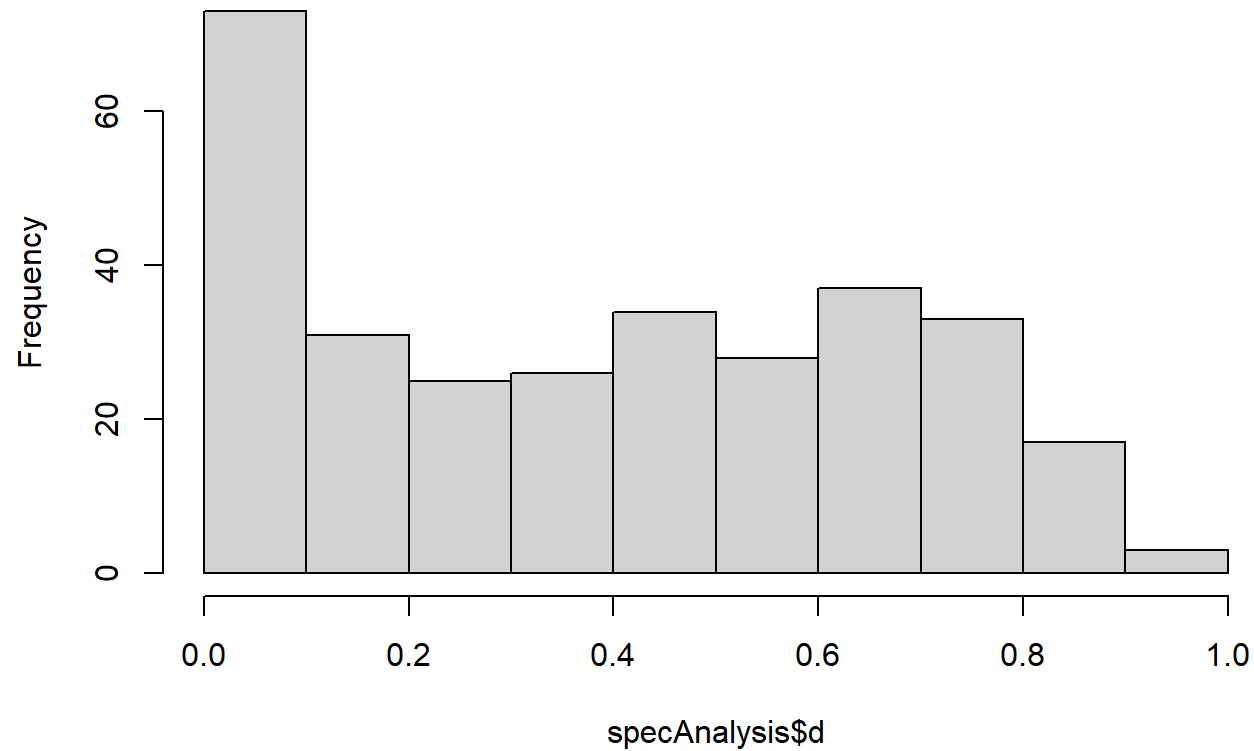
```
hist(specAnalysis$deltaAbund)
```

Histogram of specAnalysis\$deltaAbund



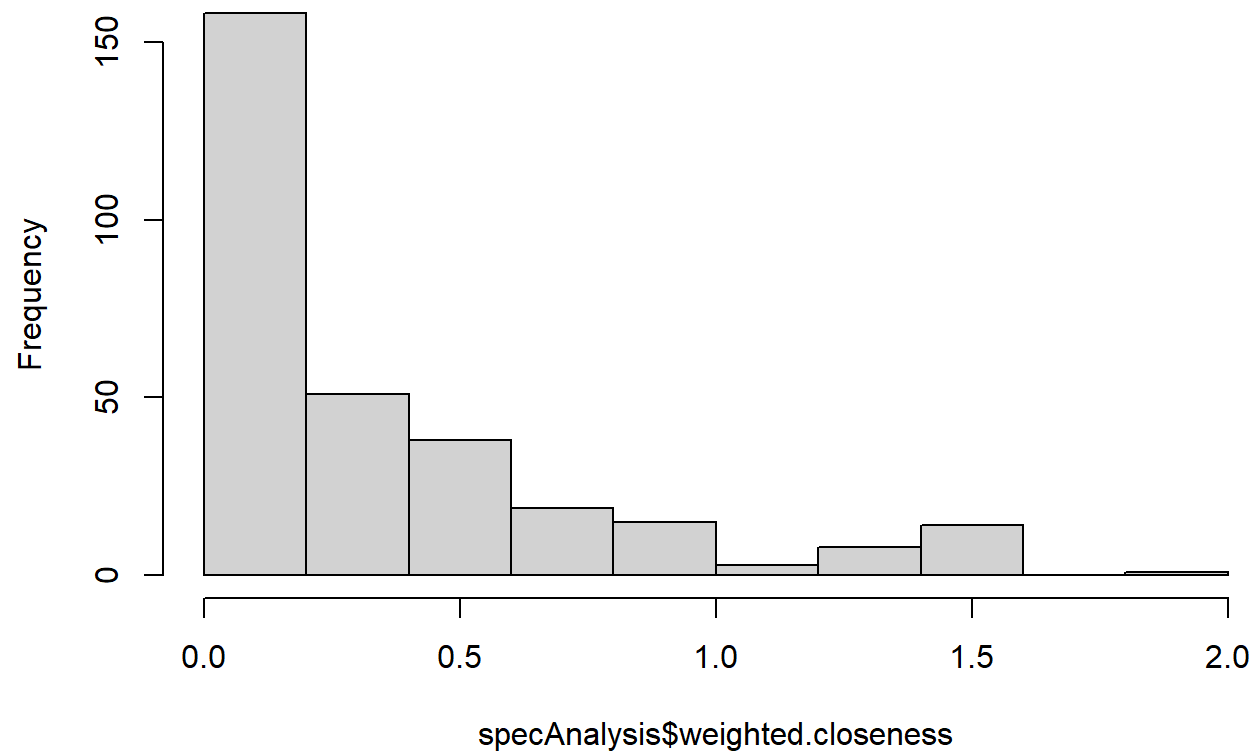
```
hist(specAnalysis$d)
```


Histogram of specAnalysis\$d



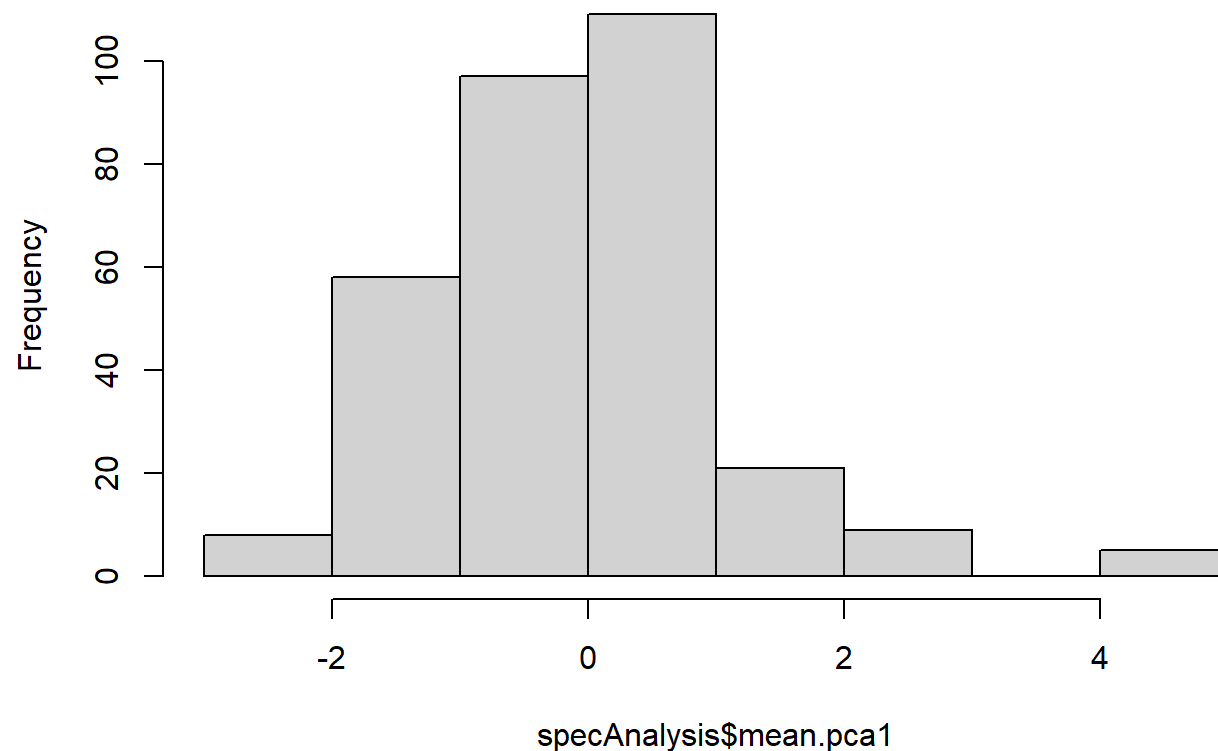
```
hist(specAnalysis$weighted.closeness)
```

Histogram of specAnalysis\$weighted.closeness



```
hist(specAnalysis$mean.pca1)
```

Histogram of specAnalysis\$mean.pca1

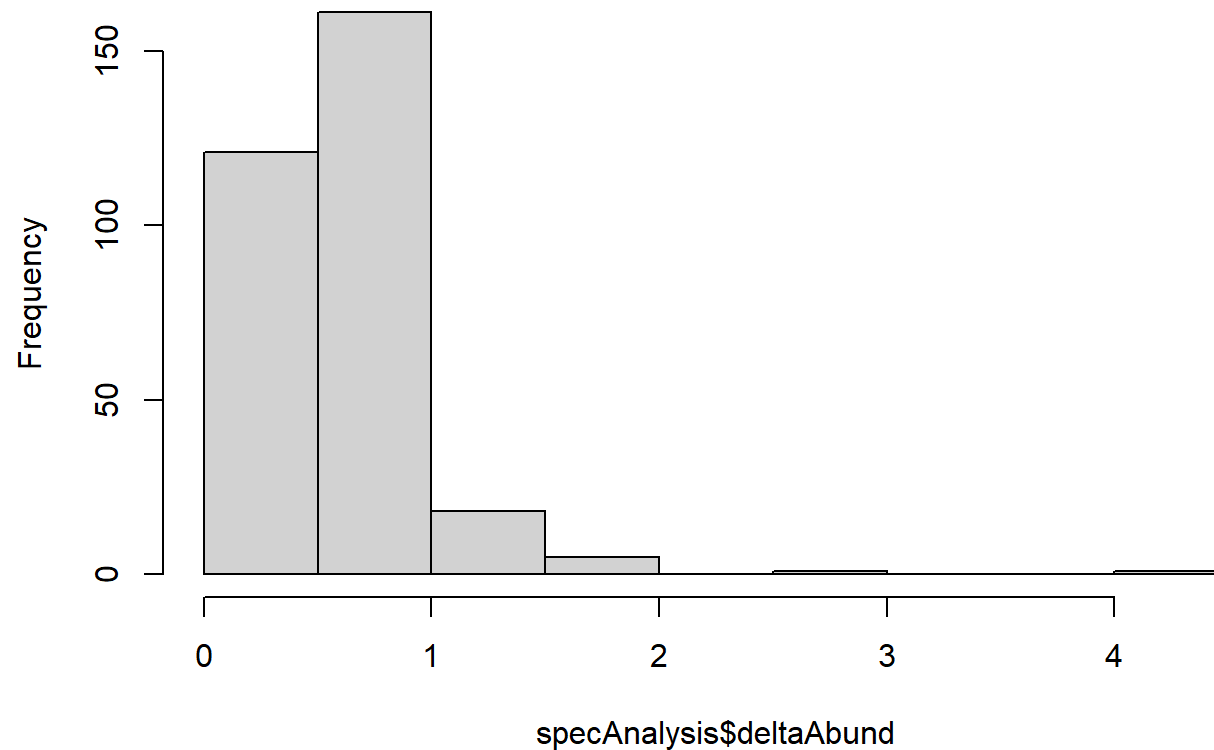


IX. Linear Models

To test my hypotheses I created linear models (6modelPlot.R). I first visualized each variable of interest using a histogram to see any extreme outliers. I created four separate linear models. Each model had a different response variable, partner variability (beta-diversity) was a fixed effect, and I included site as a random effect. I created a linear model for each of the following response variables: log-ratio floral abundance, d, nested rank, and network niche variability (i.e., using PCA variance). Example linear model: network niche variability ~ scale(beta-diversity) + (1|Site). I then used the predicted coefficients from each model to plot the results.

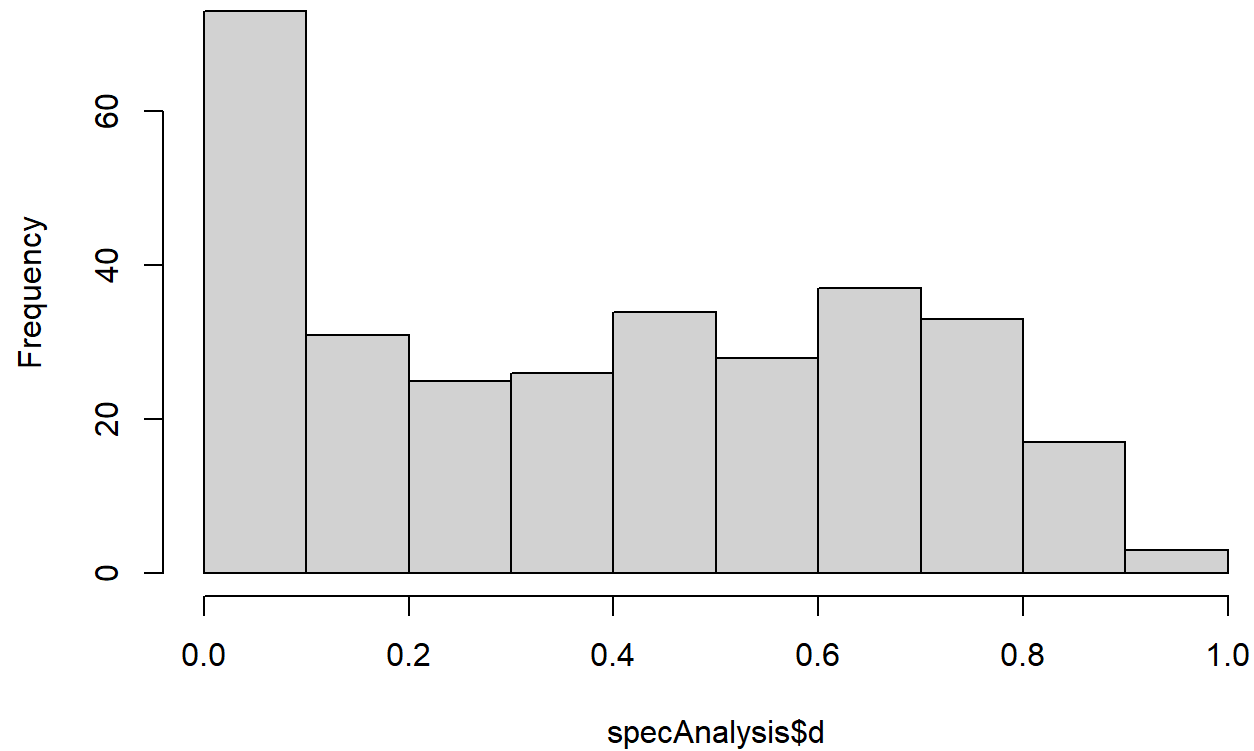
```
setwd('C:/piscagah-prairie/analysis')  
load('saved/results/specAnalysis.Rdata')  
source('src/predictIntervals.R')  
  
## visualizes and removes outliers.  
## The outliers have no effect on models and is more visually appealing when removed  
hist(specAnalysis$deltaAbund)
```

Histogram of specAnalysis\$deltaAbund



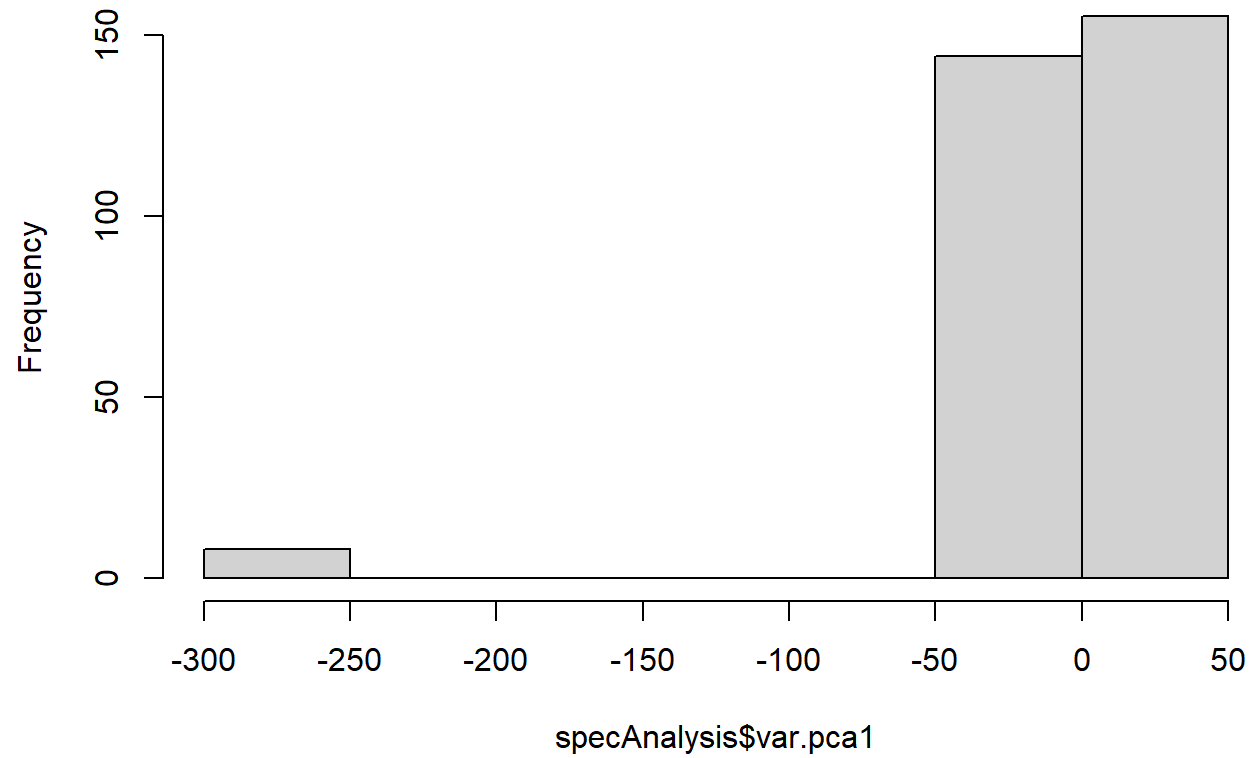
```
hist(specAnalysis$d)
```

Histogram of specAnalysis\$d



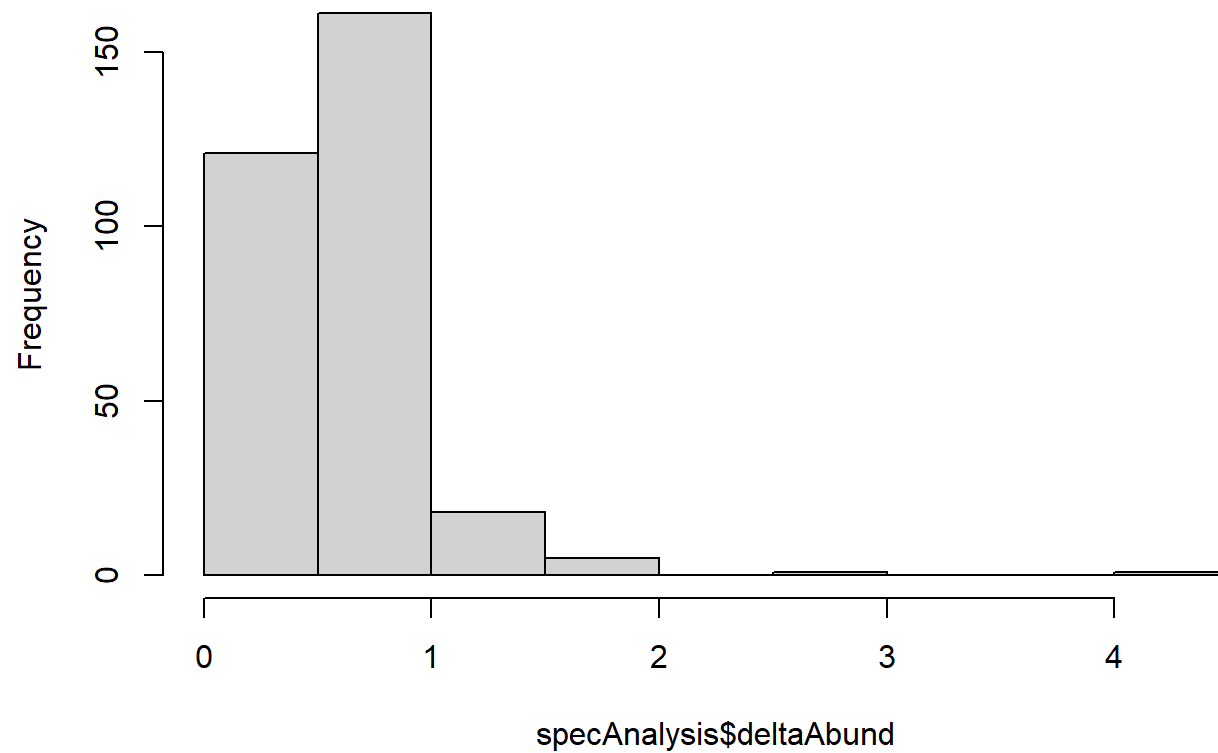
```
hist(specAnalysis$var.pca1)
```

Histogram of specAnalysis\$var.pca1



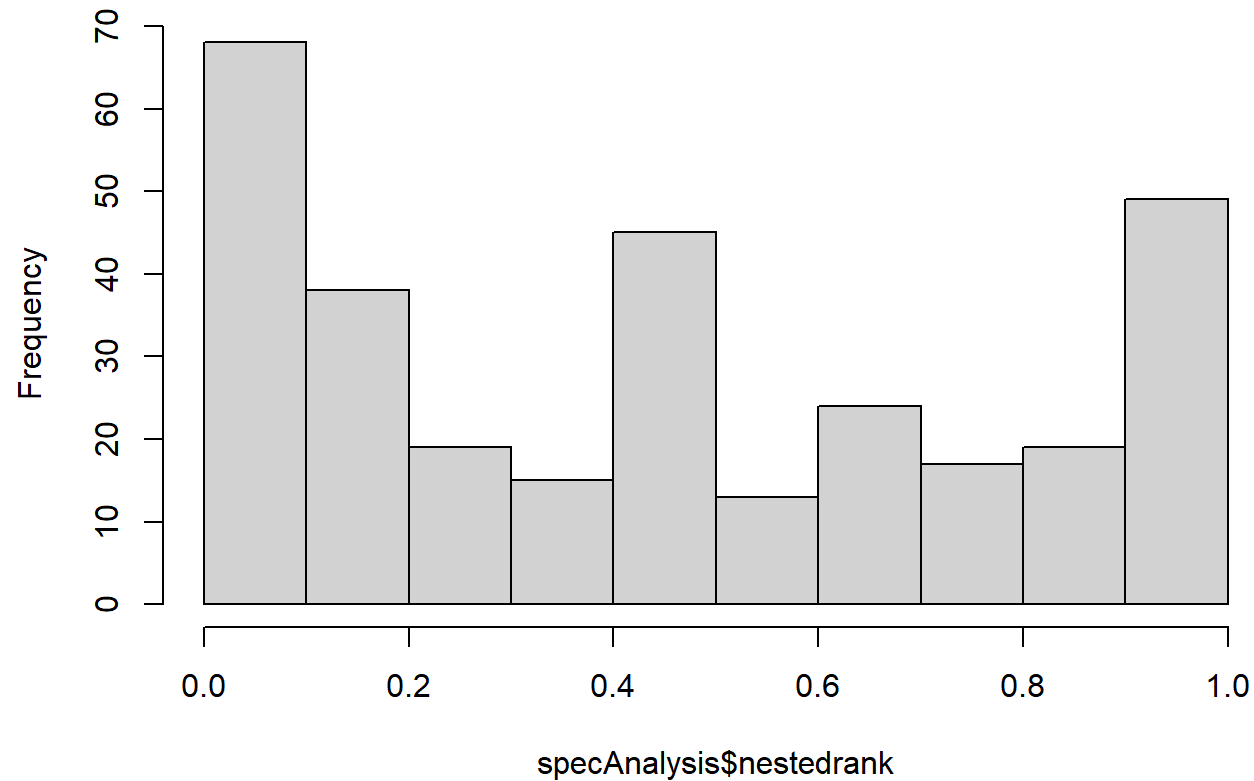
```
hist(specAnalysis$deltaAbund)
```

Histogram of specAnalysis\$deltaAbund



```
hist(specAnalysis$nestedrank)
```

Histogram of specAnalysis\$nestedrank



```
sum(is.na(specAnalysis$deltaAbund))
```

```
## [1] 0
```

```
#Create linear mixed models
#####
##Delta floral abundance vs partner variability
#####
delta.abund.mod <- lmerTest::lmer(deltaAbund ~ scale(dist) + (1|Site), data = specAnalysis)
summary(delta.abund.mod)
```



```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: deltaAbund ~ scale(dist) + (1 | Site)
## Data: specAnalysis
##
## REML criterion at convergence: 298.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.2628 -0.5368 -0.2267  0.3867  9.0457
##
## Random effects:
## Groups   Name      Variance Std.Dev.
## Site     (Intercept) 0.01828  0.1352
## Residual                0.14614  0.3823
## Number of obs: 307, groups: Site, 7
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   0.63793    0.06239   5.17981  10.224 0.000125 ***
## scale(dist)  -0.02425    0.02207  302.07175  -1.099 0.272801
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## scale(dist) -0.007
```

```
#####
##d vs partner variability
#####
d.mod <- lmerTest::lmer(d ~ scale(dist) + (1|Site), data = specAnalysis)
summary(d.mod)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: d ~ scale(dist) + (1 | Site)
## Data: specAnalysis
##
## REML criterion at convergence: 106.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.55460 -0.93354  0.06884  0.94351  1.86516
##
## Random effects:
## Groups   Name      Variance Std.Dev.
## Site     (Intercept) 0.00000  0.0000
## Residual                0.07986  0.2826
## Number of obs: 307, groups: Site, 7
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   0.38994    0.01613 305.00000  24.177  <2e-16 ***
## scale(dist)   0.01508    0.01615 305.00000   0.933   0.351
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## scale(dist) 0.000
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
#####
```

```
##Weighted closeness vs partner variability
```

```
#####
```

```
close.mod <- lmerTest::lmer(weighted.closeness ~ scale(dist) + (1|Site), data = specAnalysis)
summary(close.mod)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: weighted.closeness ~ scale(dist) + (1 | Site)
## Data: specAnalysis
##
## REML criterion at convergence: 338.6
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.0336 -0.6729 -0.3955  0.3389  3.3785
##
## Random effects:
## Groups   Name      Variance Std.Dev.
## Site     (Intercept) 0.005241 0.0724
## Residual                0.168936 0.4110
## Number of obs: 307, groups: Site, 7
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   0.37222    0.04084   2.69532   9.114 0.00423 **
## scale(dist)   0.01945    0.02367 304.06757   0.822 0.41183
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## scale(dist) 0.001
```

```
#####
```

```
##Nestedness vs partner variability
```

```
#####
```

```
nested.mod <- lmerTest::lmer(nestedrank ~ scale(dist) + (1|Site), data = specAnalysis)
summary(nested.mod)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: nestedrank ~ scale(dist) + (1 | Site)
## Data: specAnalysis
##
## REML criterion at convergence: 239.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.3684 -0.8668  0.1026  0.8237  1.6635
##
## Random effects:
## Groups   Name                Variance Std.Dev.
## Site     (Intercept)  0.000687  0.02621
## Residual                    0.123365  0.35123
## Number of obs: 307, groups: Site, 7
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   0.452063   0.023668   1.405448  19.100   0.0114 *
## scale(dist)  -0.009943   0.020137  303.904820  -0.494   0.6218
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## scale(dist) 0.003
```

```
#####
##role variability vs partner variability
#####
#identify extreme outliers and remove (-300+ variance)
which(specAnalysis$var.pca1 < -100)
```

```
## [1] 135 136 137 138 144 145 146 147
```

```
specAnalysis <- specAnalysis[-c(135, 136, 137, 138, 144, 145, 146, 147), ]
pca.mod <- lmerTest::lmer(var.pca1 ~ scale(dist) + (1|Site), data = specAnalysis)
summary(pca.mod)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: var.pca1 ~ scale(dist) + (1 | Site)
## Data: specAnalysis
##
## REML criterion at convergence: 2135.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.9612 -0.2513  0.1900  0.4358  2.9495
##
## Random effects:
## Groups   Name      Variance Std.Dev.
## Site     (Intercept) 3.006    1.734
## Residual                73.616    8.580
## Number of obs: 299, groups: Site, 7
##
## Fixed effects:
##              Estimate Std. Error    df t value Pr(>|t|)
## (Intercept)  -1.0764     0.9353   3.3188  -1.151   0.3259
## scale(dist)   1.2290     0.5009  295.6160   2.453   0.0147 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## scale(dist) 0.000
```

```
#This function uses a linear model to predict values for a network metric
#and creates plots to visualize
plot.s <- function(specAnalysis, response, mod) {
  # Create prediction grid
  expand.grid <- data.frame(
    dist = seq(min(specAnalysis$dist),
               max(specAnalysis$dist),
               length.out = 100)
  )

  # Add placeholder column for response
  expand.grid[[response]] <- 0

  # Predict using custom function
  pred <- predict.int(mod = mod,
                      dd = expand.grid,
                      y = response,
                      family = "gaussian")

  # Plot
  ggplot() +
    # geom_point(data = pred, aes(x = dist, y = .data[[response]])) +
    geom_line(data = pred, aes(x = dist, y = .data[[response]]), color = "blue", linewidth = 1.2) +
    geom_ribbon(data = pred, aes(x = dist, ymin = plo, ymax = phi), alpha = 0.2, fill = "blue") +
    theme_classic() +
    theme(text = element_text(size = 20))
}

plot.ns <- function(specAnalysis, response, mod) {
  # Create prediction grid
  expand.grid <- data.frame(
    dist = seq(min(specAnalysis$dist),
               max(specAnalysis$dist),
               length.out = 100)
  )

  # Add placeholder column for response
  expand.grid[[response]] <- 0
```

```

# Predict using custom function
pred <- predict.int(mod = mod,
                    dd = expand.grid,
                    y = response,
                    family = "gaussian")

# Plot
ggplot() +
  # geom_point(data = pred, aes(x = dist, y = .data[[response]])) +
  geom_line(data = pred, aes(x = dist, y = .data[[response]]), color = "blue", linetype = "dashed", linewidth = 1.2) +
  geom_ribbon(data = pred, aes(x = dist, ymin = plo, ymax = phi), alpha = 0.2, fill = "blue") +
  theme_classic() +
  theme(text = element_text(size = 20))
}

##Plot models
#####
##Delta abundance vs partner variability
#####
delta.abund.plot <- plot.ns(specAnalysis, "deltaAbund", delta.abund.mod)

delta.abund.plot.labeled <- delta.abund.plot + labs(x = "Partner Variability", y = "Log-ratio Floral Abundance")

#####
##d vs partner variability
#####
d.dist.plot <- plot.ns(specAnalysis, "d", d.mod)

d.dist.plot.labeled <- d.dist.plot + labs(x = "Partner Variability", y = "d")

#####
##Nestedness vs partner variability
#####
nested.dist.plot <- plot.ns(specAnalysis, "nestedrank", nested.mod)

nested.dist.plot.labeled <- nested.dist.plot + labs(x = "Partner Variability", y = "Nested Rank")

#####
##Role variability vs partner variability

```

#####

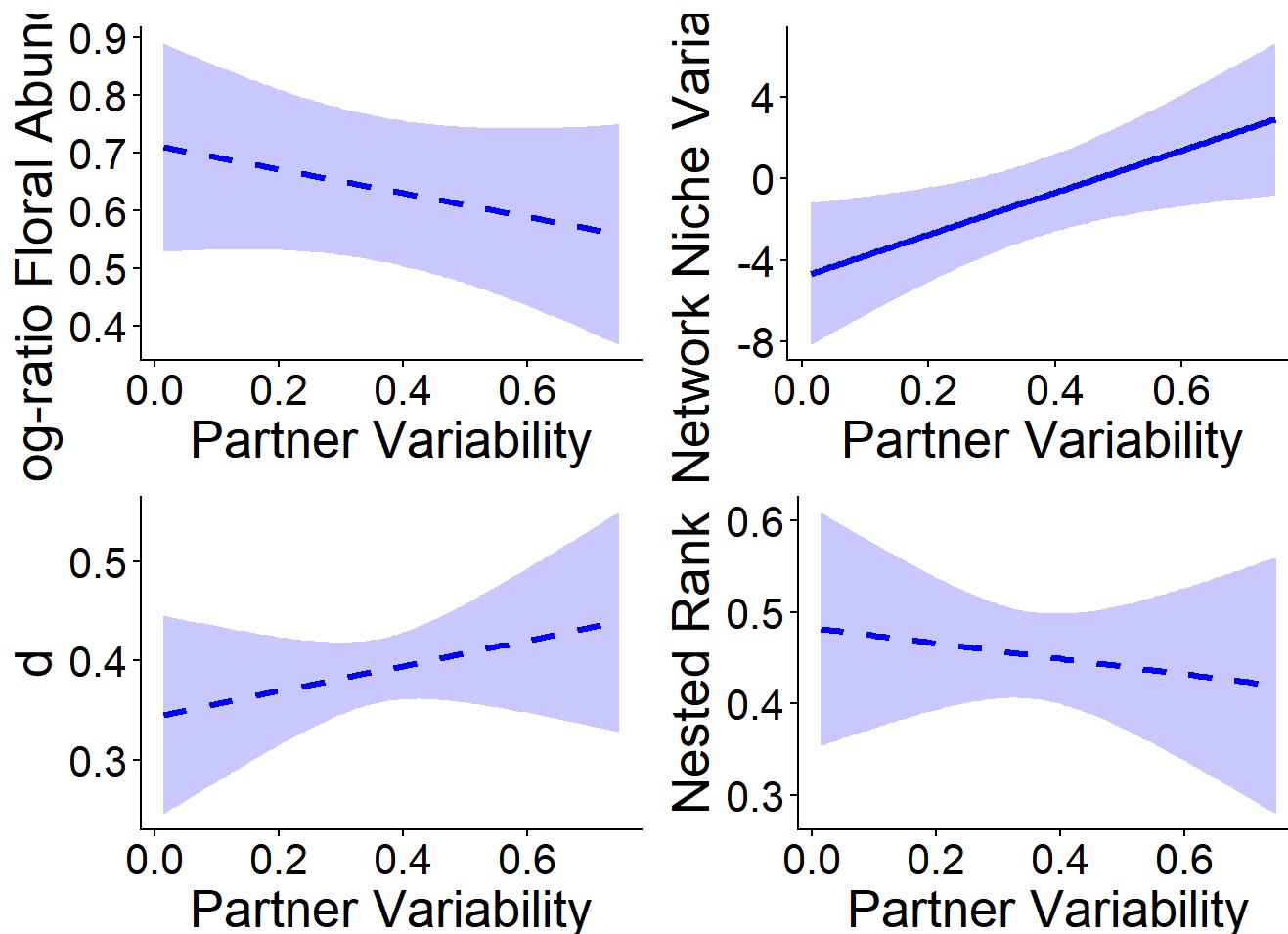
```
pca.dist.plot <- plot.s(specAnalysis, "var.pca1", pca.mod)
```

```
pca.dist.plot.labeled <- pca.dist.plot + labs(x = "Partner Variability", y = "Network Niche Variability")
```

```
#Combine plots into single grid
```

```
combinedPlot <- cowplot::plot_grid(delta.abund.plot.labeled, pca.dist.plot.labeled, d.dist.plot.labeled,  
                                   nested.dist.plot.labeled, ncol = 2)
```

```
print(combinedPlot)
```



X. Conclusions

Partner variability as a plant trait proves to be a fruitful area of study. Plants were flexible in their interaction partners across space, with comparable expected partner variability for those sites with high sampling effort. This is likely explained by communities with similar pairwise interactions locally, as the communities sampled in the present study are close in geographic area (Carstensen et al. 2014).

Plant variability was positively associated with network niche variability, whereas its associations with d and nested rank were not statistically significant. These patterns indicate that plants with flexible interaction partners may maintain similar roles within a network while also influencing the overall network structure. Gaiarsa, Kremen, and Ponisio (2021) performed a similar study from the pollinators perspective using a motif approach to quantify role variability incorporating indirect interactions. They found that role variability and partner variability were not related. I propose that differences in the structural organization of interaction partners between plants and pollinators cause each group to influence network structure in distinct ways, likely reflecting the different benefits they derive from mutualistic interactions (Van Der Kooi, Vallejo-Marín, and Leonhardt 2021).

Interestingly, partner variability was negatively associated with differences in floral abundance across pairwise years, although this relationship was not statistically significant. Floral abundance is linked to network asymmetry: abundant species tend to exert stronger interaction effects on their partners than they receive in return, whereas the opposite pattern is observed for rare species (Vázquez et al. 2007). In addition, the likelihood that a pollinator interacts with a particular plant is influenced by variation in floral abundance (Carstensen et al. 2014). In our case, from the plant's perspective, changes in pairwise interactions did not affect floral abundance.

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Altman, Bob. 2011. "Historical and Current Distribution and Populations of Bird Species in Prairie-Oak Habitats in the Pacific Northwest." *Northwest Science* 85 (2): 194–222.

Bengtsson, Jan, Torbjörn Fagerström, and Håkan Rydin. 1994. "Competition and Coexistence in Plant Communities." *Trends in Ecology & Evolution* 9 (7): 246–50.

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Cosmo, Leandro G, Ana Paula A Assis, Marcus AM de Aguiar, Mathias M Pires, Alfredo Valido, Pedro Jordano, John N Thompson, Jordi Bascompte, and Paulo R Guimaraes Jr. 2023. "Indirect Effects Shape Species Fitness in Coevolved Mutualistic Networks." *Nature* 619 (7971): 788–92.

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